6

The Dawn of Modern Culture in Europe and India

The Early-Modern Age (1300 –1800 CE)

In this chapter, we shall learn about many countries besides India. Let's first recall what we have learnt about different countries. In Class VII, we learnt that the European continent is located to the north-west of India. You already know a few important countries in Europe. Locate the following countries on the map of Europe in your atlas or on the world map in your class:

1. The country that ruled India before 1947.
2. The country whose capital is Paris.
3. A country with an ancient civilisation where famous philosophers such as Socrates and Plato lived.
4. A country that is surrounded on three sides by the Mediterranean Sea.
5. A developed country in central Europe whose capital is Berlin.
6. A huge country in Eastern Europe whose capital is Moscow.
7. If we travel by land from India to Paris, which are the countries we will cross?
8. If we want to go to England by sea, trace the route we will have to take on the map with your finger.
9. Locate Iran on the map. This country has very ancient cultural ties with India.
10. Locate Iraq to the west of Iran – this country had a civilisation contemporary to Harappa.
11. Locate the Arabian Peninsula. Islam was born here and Christianity began in Palestine.

The Modern Age and Before

There are many characteristics of the modern age we live in. What do you think are the special features of the modern age? Are they different from the earlier ages? Are the main differences in the economic system, the social system, in culture and ideas, or in the political system? Discuss in your class.

Historians also debate issues like 'What are the special features of the modern age?' – just like you did in class. It is widely accepted that the two main features that distinguish the modern age from earlier eras are industrial production and democratic governments. It is more difficult to pinpoint the social and cultural changes that have taken place in the modern age but we can still make some general observations – we give less importance to a person's caste background now, everyone is

treated equally before the law, and the gap between the rich and poor is wider now than it was before. In earlier times, religion played a bigger role in people's lives – they were more influenced by their religious beliefs. Today, people have more faith in science and reason.

In this chapter we shall trace how and when the society of the middle ages moved into the modern era. The age when this process began is called the Early-Modern era.

The process of change in Asia and Europe began sometime from the 14th century (after 1300 CE) and continued till 1750 CE. Religion, politics, art, commerce and trade saw many changes during these 450 years. More and more people began to live in cities and they also began to think differently. This pre-modern age laid the ground for the industrial revolution and the French revolution.

Common Era: While each community or region has its own way of calculating time, today all countries the world over have adopted a common method of keeping time and dates. This is called the Common Era (or CE in short). We are living in the 21st century of the Common Era. The time before the beginning of the Common Era is called Before the Common Era (BCE in short).

What developments took place in the Asian countries like India, Iraq, Iran and Turkey and the European countries like Italy, Holland, France and England between the 14th and 18th century that led to the economic and political revolutions? The changes were many, but we shall discuss only the cultural, religious and intellectual changes that took place.

Three important processes occurred between the 14th and 18th century: they are called the Renaissance, the Reformation and the Enlightenment. These movements are associated with European history but similar developments took place in the Asian countries during this period. For example, in India, we had the Bhakti and Sufi movements and the new awakening in art that happened during this time. In fact, Asia and Africa contributed significantly to many of the changes that occurred in Europe. That's why, since the past few decades, historians have been saying that we must study the changes not just in the European context but on a global scale.

First, locate India, Iran, Iraq, Turkey, Spain, Italy, France, Holland, England, etc. in your atlas or on the wall map in your class.

Then, identify some of the important towns in these countries. For example, Vijaynagara, Surat, Delhi, Tehran, Isfahan, Baghdad, Istanbul, Venice, Rome, Florence, Geneva, Paris, London, etc.

6.1 The Different Facets of Change

What exactly happened between 1300 and 1750 CE? Let's find out how the modern era began.

Trade and urbanisation: During this period, trade expanded rapidly between Europe, North Africa and Asia. Commerce expanded within countries as well, with different kinds of goods being traded. The growth of trade also saw the growth of cities in the three continents. New cities were born and older cities expanded. This urbanisation brought with it many changes. A new class of wealthy merchants emerged. People travelled more between cities, leading to an exchange and spread of new ideas, inventions and technologies.

There was one striking difference between the cities of India and Europe. Many of the cities in Europe were not ruled by kings. They were independent and autonomous and were governed by

organisations formed by merchants and artisans. So, to some extent, democratic and republican ideas began to take root in these cities. These cities included Genova and Florence in Italy, Flanders in Holland, etc. Big cities like Surat, Agra and Vijayanagara also developed in India at this time, but they were still under the rule of kings or the control of feudatory lords.

What kind of goods and ideas do you think were exchanged during the 14th century?

Centralised states: This was the time when many powerful kingdoms were established. For example, in India, we had the Vijayanagara Empire and the Mughal dynasty. Iran had the Safvid Empire and Turkey the Ottomans. Many strong kingdoms were established in France, Spain and England as well, apart from several smaller kingdoms in Italy.

These kings were all very ambitious so they kept a tight control on their power over their kingdoms. As a result, power, wealth and authority gradually began to get concentrated in their hands. We call these kingdoms centralised kingdoms because economic and political power was centralised in the king. Before this time, zamindars (big landlords) and feudal lords used to rule over their lands and estates and govern their subjects. Now their powers were curtailed or completely taken over.

Many people say that the government today has more power and authority than the centralised kingdom of the Mughals. Do you agree with this viewpoint? Give reasons for your answer.

The emergence of an urban middle class: The expansion of trade, the growth of cities and the centralisation of all authority under the king led to one important change - the emergence of a new middle class in these countries. It consisted of merchants, rich artisans, accountants, clerks, lawyers, professional artists, poets, writers and so on. Urbanisation and development also saw new kinds of jobs being created (for example, book-keeping and accountancy, administration, tax collection, judicial and legal work, diplomatic contacts between kingdoms, etc). The people doing these jobs became part of the middle class. This class was always trying to earn more money and hence prepared to go to distant kingdoms in search of good jobs. So there was constant movement of people across a large geographical area.

This wide exposure made the middle class look at everything happening around them with a more critical eye. The people were not afraid of criticising their religious leaders or their rulers. Education played a big role in shaping them. To be educated was not just becoming literate. People needed to have a wide grasp of literary knowledge to perform their jobs well.

In Europe, education meant studying Greek and Latin classics, while in India it meant studying Persian and Sanskrit classics. But there was also wide exchange of knowledge. The Greek texts of Aristotle and Plato and Indian classics such as the Panchtantra and mathematical treatises were translated into different languages and widely read. During the reign of the Mughals in India, many Sanskrit classics such as the Ramayana, Mahabharatha, Upanishads etc were translated into Arabic and Persian.

Though there were many similarities between the middle class in Europe and India, some important differences need to be noted. First, this class in India was limited to a few specific caste groups, for example, Kayastha, Kshatriya and Brahmin. In Europe, people from different occupational groups were included into this class. The second significant difference was that the middle class in India showed no special interest in science and mathematics, whereas these areas of knowledge were important in Europe.

6.2 The Renaissance

6.2.1 The Study of Ancient Classics and Travel Abroad

Most of the Latin and Greek classics focused on humanity and human life. So the main areas of study and inquiry were politics, ethics, philosophy, law, civility, nature and the physical world, etc. Since this study was focused on human beings, it was known as humanism, or the humanities. It was different from religious thought. It was not based on believing any religious text or religious leader or on making sacrifices in this world for heavenly salvation. It did not talk about penance or suffering. What it emphasised was rational and critical thinking. It encouraged people to think for themselves and not blindly accept what anyone said. So even the teachings of religious leaders were examined rationally and critically.

The Middle Ages was the time when Christianity had established itself across Europe. The powerful Christian church did not like its teachings to be questioned. It said the Greek and Latin classics were written before the birth of Christ, so they could not be in harmony with the Christian scriptures. The church preached that people should not pursue worldly wealth and prosperity but should seek salvation in heaven. As a result, few people studied the Greek and Latin classics during the Middle Ages. These texts gradually disappeared from Western Europe because of the church's opposition.

However, the Islamic kingdoms of Iran, Iraq etc continued to translate and study the classics. Islamic culture flourished in the 14th century, its influence spreading from India to Spain. During this time, Islamic scholars translated the literary classics of China, India, Iran and Greece into Arabic and Persian. So the entire Islamic region From Asia to Europe was exposed to a mix of Chinese, Central Asian, Indian, Iranian, Iraqi, Egyptian, Greek and other cultures.

When interest in the Greek and Latin classics revived in Western Europe in the 14th century, the classical texts preserved in the Arabic translations came in handy. Also, ancient mathematical and astronomical knowledge from India and scientific knowledge from China found their way to European scholars.

What did the church think of the Greek and Latin classics? Why did it think this way?

What were the subjects covered in the ancient Greek classics? Make a list.

6.1: Francisco Petrarch

6.2.2 Humanism in Europe

European scholars began taking an interest in studying Latin and Greek texts around 1300 CE. Expanding trade, urbanisation and the rise of powerful monarchies were creating new jobs. They wanted to explore these challenges and take advantage of the new professions and jobs.

What new professions and work do you think urbanisation and the formation of powerful new monarchies could have created?

What do you think were the problems people faced with kings who wanted to become powerful and merchants who wanted to earn profits?

The Latin scholar Francisco Petrarch (born 1304, died 1374 CE) of Italy is considered to be the person who ushered in the humanist movement. He was distressed by the fact that people at that time were not using language properly. He began studying Latin classics to understand more about language and its usage. He found that the ancient texts not only helped him to learn how to use language correctly but also how to think critically to understand the world better.

The study of Latin and Greek began to spread rapidly under the influence of Petrarch and other scholars. The humanists believed that learning these texts could improve the ability of young people to think, write formally, address the public, argue a case in court, or talk intelligently about trade and politics. So they began to establish schools to impart classical education. They were helped in their task by the invention of the printing press in 1439 by the German artisan Johannes Gutenberg (born 1395, died 1468 CE). Hundreds of copies of books could now be easily printed. Scholars far and wide could lay their hands on printed copies of old and new books. This stimulated debate and the exchange of ideas between people.

Figure 6.2: A portrait of Erasmus dated 1526 by the German artist Albrecht Dürer. This portrait was printed and widely distributed. What impression does this portrait create about Erasmus? Where and how are the names of Erasmus and Durer written? How are the numerals 1526 depicted?

While printing gave an impetus to the study of Greek and Latin texts, the scholars chose to write in the regional languages they knew, like Italian, German, English, Flemish and French. They wanted to write in a language that the common people could easily understand.

What was the basic difference between humanist study during the Renaissance and learning during the Middle Ages? We saw earlier that in the Middle Ages learning focused on religious themes. No scholar had the courage to write or say anything that went against what the church thought or said. But humanist study focused on the daily life of people, like the love story of a hero and heroine, the political system, or socio-economic issues.

The humanists no longer hesitated to air their views about the church and to even criticise it, even if they depended on the church for their living. For example, the Latin scholar Lorenzo Valla studied some important church documents in 1435 and proclaimed they were forgeries. One such document with the seal of a Roman emperor granted temporal power to the Roman Catholic Church.

The well-known Dutch humanist Erasmus (born 1466, died 1536 CE) studied the writings of early

Greek Christians as well as the original Greek Bible. He pointed out that the church's translation of the Bible into Latin had many errors. He also claimed that many of the beliefs of the church verged on superstition. He wrote a satirical book, 'In Praise of Folly', in which he took a critical look at many of the church's views and actions.

Did women also participate in humanist studies? In those days, only men received formal education. Women were expected to look after the household. But there were a few women who broke these barriers and studied Greek and Latin, making a name for themselves among the humanist writers. One of them was Cassandra Fedele (born 1465, died 1558 CE). She believed that women, too, should be allowed to learn the classics. In those days, Venice was a republic but women weren't allowed to participate in public life. Fedele felt this was a barrier that limited freedom and democracy and gave precedence to the needs and wishes of men. She was among the first women to criticise the male-centric system, paving the way for later feminist thinking to emerge.

Another famed humanist was the Italian Niccolo Machiavelli (born 1469, died 1527 CE). He published an important book called 'The Prince' in 1513. It was basically a study of politics. It investigated how political processes actually happen in the real world rather than some imaginary or ideal situation. It describes how a king can acquire autocratic power by freeing himself of all moral compunctions.

In this way, we see how the study of ancient classics gave rise to humanism. Humanist studies strengthened people's powers of self-expression and critical thinking. It eventually led to opposition of the church. Among the permanent changes it made, the most important were liberal classical education and intellectual freedom.

Make a list of the most prominent features of humanism.

Who is considered to be the originator of the humanist movement? What was his main concern?

Did women have a role in humanism?

What was the subject of Machiavelli's book?

How did the invention of the printing press help to spread humanism?

Do you think intellectuals should have unlimited freedom to criticise society or should the government set limits to their freedom? Explain with reasons.

Why did European intellectuals study the ancient classics in their search for new ideas?

What impact did the printing of books have on intellectual freedom?

Today, a new technique is replacing printing. What is this technique? What impact is it having on intellectual freedom?

In India

We saw earlier that even in India, Sanskrit, Persian, Arabic and other classics were widely studied and translated. The Mughal emperors and even the minor kings encouraged such initiatives. This was the age when the *munshis* and *munims* became indispensable. No ruler could administer his kingdom without these educated managers and accountants. They wrote many books about the Mughal period. Their analytical skills and descriptive powers are reflected in these treatises.

The middle class during the Mughal period had one more striking feature. It represented a mix of

Indian and Central Asian cultures. The people took inspiration from all these different sources. This influenced the way they lived and the lifestyle they adopted. That was not all. Though they depended on emperors and kings for their livelihood, they were proud of their independence, even criticising them. For example, many munshis even criticised Aurangzeb for his religious policies.

There was one other significant difference between India and Europe. In Europe, the printing press was widely used to publish books and spread ideas. In India, the use of this new invention was next to negligible, which hampered the spread of new thinking.

The *munshi* and *munim*

During Mughal times, these people worked in the rulers' courts and with merchants. They performed administrative tasks such as correspondence, preparing legal documents, maintaining the books of accounts, writing about various events, etc. They were familiar with the practices, customs and languages of several countries.

Around 1689, a writer from Chattisgarh's Ratanpur kingdom wrote a satirical poem *'Khub Tamasha'* in which he poked fun at the kings and queens of Ratanpur and their diwans. His work clearly reflects mixed cultural influences - the two words in the title are of Persian origin.

6.2.3 A New Artistic Trands

One other change in India and several other countries was the flowering of the creative arts after 1300. The most visible impact of this new wave of creativity was seen in architecture. When the Turks came to India, they brought with them new building techniques, such as the use of domes and arches. This technique of using domes and arches in buildings was first developed in ancient Greece and Rome.

Figure 6.3 shows pillars supporting the weight of the roof of a Greek temple. In India too pillars and beams were used to support

Figure 6.3: A Greek temple

Figure 6.4: The Kandariya Mahadeva

the roofs. Massive temples were built in India using pillars and horizontal beams. You can see an example of this in figure.6.4 of Kandariya Mahadev Temple in Khajuraho which was built about a thousand years ago. This splendid example of temple architecture combined hundreds of small *shikaharas* or spires to form an impressive edifice.

Figure 6.5: The Notre Dame Cathedral in Paris, France

Figure 6.6: A Roman arch

In Northern Europe, churches with similar complex structures were built during the 12th century. One excellent example is the famed Notre Dame Cathedral in Paris, which is shown in Figure 6.5. This architectural style is called the 'Gothic' style.

Figure 6.6 also shows arches used in the Roman Empire 2,000 years ago to support the weight of the roof of structures. Arches were used in constructing domes as well. One example is the dome of the Pantheon (Figure 6.7), built 2,000 years ago in Rome. It looks absolutely grand when seen from the inside but it isn't as impressive when seen from outside (compare Figure 6.7 with Figure 6.8).

Figure 6.7: An inside view of the Pantheon

Domes that look impressive externally were the contribution of Islamic architecture. See Figure 6.9, which shows the 'Dome of the Rock' built in Jerusalem in 691 CE. We can see how grand the dome looks above an equally impressive base composed of beautifully arranged arches.

In India, the use of domes and arches made a very attractive mix with elements of temple architecture in the early modern period. As a result of this fusion, a new and transformed Indian architecture took shape during this period, which both Hindu

Figure 6.8: An external view of the Pantheon

Figure 6.9: The Dome of the Rock, Jerusalem in (691 CE)

and Muslim rulers adopted. We can see examples of this architectural form in India in the Lotus Mahal of Vijaynagar, the palaces of Fatehpur Sikri and so on. The domes and arches of the Taj Mahal (1648 CE) beautifully incorporate Indian design and engraving - for example, the inverted lotus crowning the dome and the brass finial emanating from it in Figure 6.10. The dome is shaped beautifully, round like a teardrop.

Figure 6.10: The Taj Mahal

Thus, the architectural journey of domes and arches began in the ancient Roman Empire. They were almost forgotten by Western Europe of the Middle Ages. They then began to be used extensively in the architecture of Islamic countries and India. In the process, they took on new forms and styles.

Figure 6.11: St Peter's Cathedral in Rome designed by Michelangelo in 1547

In Europe, interest in ancient Greek and Roman culture was revived in the 14th century, beginning with Italy. Architects began using domes and arches once again. The style they adopted was developed by Islamic architecture. When Michelangelo designed the St Peter's Cathedral in 1547 CE, he took inspiration from Greek temples and Roman and Islamic domes.

Trace the Greek, Roman and Islamic influences in Figure 6.11. St Peters Cathedral is considered to be one of the finest examples of Renaissance architecture. Compare it with the Notre Dame Cathedral. What do you think are the differences in the vision behind these two structures?

This architectural style in which tall Greek columns, arches and domes are used is called the 'classical style'. It continues to be used even today.

6.2.4 Painting and Sculpture

The Renaissance is most renowned for its magnificent paintings and sculpture. In Europe of the Middle Ages, painting had limited scope – restricted to Biblical figures - as seen in Figure 6.12. Things

Figure 6.12: A 13th century painting of mother Mary, infant Jesus and saints

began changing towards the end of the 13th century, when artists began painting based on their observations of real humans. Religious figures continued to dominate but the elite and rich, successful professionals began to get their portraits painted. Even the religious figures became more life-like, depicting different emotions, forms and perspectives.

Figure 6.13: 'The Marriage of the Virgin' – painted by Raphael around 1500

Colour, Technique and Science: With new colours being introduced from India and Iran, artists began experimenting with colour and methods. They also used new oil-based colours. They mixed colours to get innumerable shades.

Their paintings also became more realistic as they began depicting perfect replicas of what they saw. This became possible with an understanding of perspective. When we look at a scene, objects that are close look bigger while objects that are faraway look smaller. Objects in the background become smaller in geometric proportion. Calculating this proportion helped artists to separate near and distant objects in their paintings.

One famous example is seen in the painting in Figure 6.13 by the famous artist Raphael (born 1483, died 1520 CE). We can distinguish close from far by seeing small human figures in the distance while the figures in the foreground are bigger. We can see the lines linking the figure in the front and the building in the background converging This gives an idea of geometric proportion.

Another technique emerged from science, which was developing rapidly during this period. The physician Andreas Vesalius dissected human cadavers to study the structure of bones, muscles and other internal organs and published books on his findings. Artists could now study human anatomy to perfect their human figures. Many of them even dissected human bodies to get a better idea of the internal structure. The most famous among them was Leonardo da Vinci.

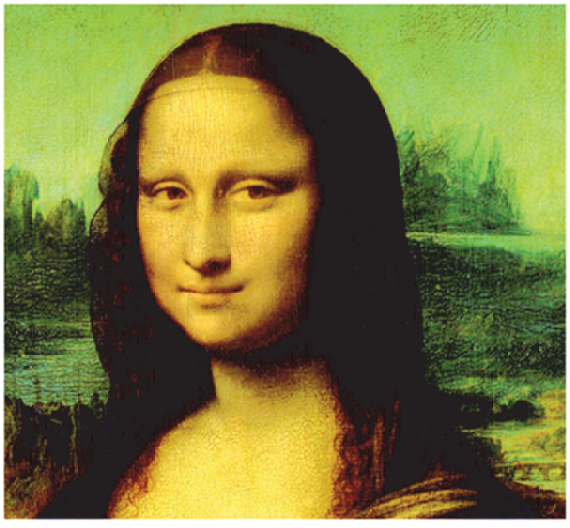

Figure 6.14: Leonardo da Vinci's 'Mona Lisa'

Two Important Painters

Italy had many great painters during the Renaissance period but we shall discuss the two most important among them. They were Leonardo da Vinci (born 1452, died 1519 CE) and Michelangelo (born 1475, died 1564 CE).

Da Vinci was a scientist, sculptor, architect, inventor and painter. He was the perfect example of the 'Renaissance Man', with his many-sided personality and wide range of interests. His most widely discussed painting is the 'Monalisa'. It depicts a woman with a captivating smile (Figure 6.14). Scholars have tried to decipher the mystery of that smile for the past 500 years. In a way, it was symbolic of that age. Never before had any artist given such prominence to a momentary and fleeting facial expression, capturing it for eternity in a painting.

Figure 6.15: 'The Creation of the Sun, Moon and Vegetation' – one of the Biblical scenes painted by Michelangelo on the roof of the Sistine Chapel. Here god is shown creating the sun. The intensity of creation is reflected in his facial expression and sharp gaze, which dazzles even the angels

Michelangelo is counted among the most prominent artists in world history. He lived in the city of Florence. He was under the patronage of the Pope for whom he undertook many creative projects. The most illustrious of these was painting scenes from the Bible on the walls and roof of the Sistine Chapel. You can see a portion of one of the paintings in Figure 6.15.

Michelangelo was as good a sculptor as he was a painter. He used Italy's white marble to sculpt extraordinary figures. Take a look at the most famous among them – 'La Pieta'. The portrayal of the human body, emotions and even clothes is spectacularly realistic. The grief of the mother with the dead body of her son in her lap is deeply and profoundly etched (Figure 6.16 and 6.17).

Much of Renaissance art was inspired by Christianity. But, at the same time, we see the marked influence of ancient Roman art. Artists like Michelangelo studied ancient paintings, sculptures and buildings in minute detail and tried to replicate their style.

Figure 6.16: 'La Pieta' – a sculpture by Michelangelo, showing Virgin Mary holding the body of her son Jesus Christ

Figure 6.17: A close-up of Mary's grief-stricken face

Early-modern Painting in India

A new chapter opened in Indian painting in the 15th century. This was the age of miniature paintings in which bright colours were used to paint pictures on paper. These miniatures had two sources of inspiration. One was the simple illustrations on palm leaf handwritten books. Using deep and bright colours, they depicted idealized – not realistic - human and animal figures (Figure 6.18).

The second source of inspiration was the miniature tradition of Iran. The country had a practice of producing books handwritten on paper with illustrations of the incidents described in them. This was around the time the Renaissance began in Europe. The most famous Iranian painter was Kamaleddin

Figure 6.18: An illustration in a Gujarati Jain manuscript from the 14th century. The father of Lord Mahavira is seen talking to some person. Examine the colours used and how the human figures are depicted

Figure 6.19: 'A class in a madrasa' – a miniature painting by the Iranian artist Behzad. Despite its miniature size, the artist has depicted minute details of the people, surroundings and buildings. He has also treated the natural and architectural components of the scene in a very imaginative way. However, the figures in the foreground and background are of the same size..

Behzad (born 1450, died 1535 CE). One of his paintings is shown in Figure 6.19. When the Mughal dynasty was established in India, the emperors invited many Iranian artists to the country. Traditional Indian painters who were skilled in using a wide range of bright colours in their paintings worked with these Iranian artists and together they produced a really wonderful new tradition of painting called 'Mughal Miniature Painting'.

Figure 6.20 is a painting done by Miskin, a Mughal court painter during the reign of Emperor Akbar. This painting depicts a scene from the Mahabharata and its appendix Harivansh. Shri Krishna is seen

Figure 6.20: Miskin's miniature painting of Shri Krishna holding the Govardhana Hill.

lifting the Govardhana Hill. The cows, their herders and other people are sheltering beneath. This complex composition has many distinct individual figures. These people are not victorious warriors or rich merchants but ordinary village folk.

Now look at the painting in Figure 6.21. It was done by a painter called Bichitra around 1620, during the reign of Emperor Jahangir. It is a symbolic painting that depicts the emperor's spiritual leanings. Jahangir is shown sitting on an hour-glass throne depicting the flow of time. A sun-like halo crowns him, with European-style angels on both sides. Jahangir is presenting a book to a Sufi saint. The Ottoman Sultan, the King of England and Bichitra himself are shown standing nearby. In this painting Jahangir wants to show himself as an emperor who has mastered time and who cares for religious leaders more than emperors and kings. Despite the exaggerations and symbols used in his painting, Bichitra still depicts human figures realistically, giving each of them a distinct and individual look.

Do you think artists and painters should always paint realistic portraits? Give reasons for your answer.

Have you seen a painting or sculpture done by a famous artist? If yes, then tell the class about it. Was the painter a realistic artist? What impact did the painting leave on you?

Why do you think the new and powerful kings and emperors, the pope, wealthy merchants and others preferred and encouraged realistic art?

6.3 The Scientific Revolution

Theism – the belief in god(s) – dominated thinking in Europe during the Middle Ages. It was also the dominant influence in the Islamic countries and India. The theists preached that god is the ultimate truth, the all-powerful who created the universe. Hence, people should forsake the pursuit of comfort and prosperity in this world and seek a place near god in heaven. There was no place for studying the

physical world in such thinking. Such study was even seen as being opposed to religious life. This is why science could not progress in the Middle Ages. But this does not mean there was no science. The Islamic world saw a flowering of science during this period.

We had seen earlier how Arab philosophers studied and translated the work of Chinese scientists and Indian mathematicians. They adopted Indian mathematics, particularly the decimal number system. They also adopted some well-known Chinese inventions such as gunpowder, block printing and magnets. Arab intellectuals built upon the work of Chinese and Indian astronomers in calculating the motion of planets, stars and other heavenly bodies. The scientific treatises from these countries found their way to European scientists. In this way, Arab scientists reached Indian and Chinese science to Europe.

Figure 6.21: A painting of Jahangir with contemporary kings by Bichitra. The artist had never seen the kings of Turkey or England so how do you think he painted their portraits?

When the study of Greek and Latin classics was revived in Europe, the people there began studying scientific literature from Greece as well as the works of Arab science. It was in those days that the European navigators were looking for a sea route to India and China. They had to cross unfamiliar seas and oceans. Their only method to navigate their way across the ocean was by tracking the position of stars and planets in the sky. They also used magnetic compasses to chart their path. The telescope with glass lens had just been invented so it helped them to look long distances across the sea.

Similarly, in those days cannons were used widely in war. Cannoneers had to know the angle at which the cannon should release the cannonball for it to strike the target. The estimate of the trajectory included factors like the weight of the cannonball and the bore diameter of the cannon. All this involved scientific study. That's why kings and merchants showed special interest in science and its investigations.

There were some scientists in Europe during the Middle Ages. In 13th century England, there was Roger Bacon (born 1220, died 1292 CE) who paved the way for experiment-based investigations of scientific principles. In those days, the common practice was to guess or estimate the answer to any question. The process of reaching conclusions on the basis of observations from experiments was not established. Bacon's insistence on this scientific method slowly became the accepted practice in scientific studies.

Figure 6.22: A portrait of Nicolas Copernicus

As we had seen earlier, the Renaissance was the age when a lot of research was being done on the anatomy of the human body. Another area of study was the geometry of perspective or projective geometry. In addition, there was the observation based astronomy. The widely accepted belief during the Middle Ages was that the earth was the centre of the universe and all the stars, sun, moon and planets orbited around it. This geocentric theory can be traced to the ancient Greek philosopher Aristotle. People in those days also believed that the earth was flat and that we will fall down if we went to its edge.

Many astronomers of the time studied the path and motion of planets in the sky. They faced a problem. They found that the motion of the planets did not tally with a geocentric universe. The most prominent among them was the Polish astronomer Nikolas Copernicus (born 1473, died 1543 CE). He published a book towards the end of his life in which he said that this problem could be resolved if we accepted that it is the sun, not the earth, that is at the centre of the universe and that all the planets, including the earth, revolved around it.

This heliocentric theory of Copernicus was quite revolutionary for his time. People were not ready to accept that the earth, which appeared stationary to them, could actually be orbiting the sun. The church stated that god had ordained that the earth was the centre of the universe so that humans could live at the centre. So the church opposed the heliocentric theory.

However, many astronomers investigated Copernicus' theory. They made detailed and minute observations of the motion of stars and planets in the sky. Prominent among them was Tycho Brahe (born 1546, died 1601 CE), who set up an observatory in Denmark. Here, new methods were used to observe the planets and calculate their motion. Another astronomer, Johannes Kepler (born 1571, died 1630 CE), examined these findings and made his own detailed calculations and observations. He noted that the calculations made sense only if the earth and all the planets revolved around the sun, as Copernicus had said.

However, Copernicus had erred in one respect. He said the earth followed a circular path while orbiting the sun. Kepler's calculations showed the orbital path followed by the earth wasn't circular but elliptical.

It was Kepler who said we can understand the physical world only by minute observations and calculations, not by our widely held beliefs, religious texts or assumptions. The Italian scientist Galileo Galilei (born 1564, died 1642 CE) also placed the greatest emphasis on detailed observations and calculations. He established what Kepler said on firm ground. Galileo is known for asking leading questions, then looking for answers by doing experiments and studying the results. These became the guiding principles of scientific investigations.

Galileo used the telescope invented by seafarers to study the stars. He found that planets like Jupiter and Saturn had several moons orbiting around them. The question he asked was: Why do planets and moons orbit? Who or what propels them?

He did many experiments to find out why objects move and why they stop. For example, he dropped heavy and light objects together from a height to see if they fell together or if there was any difference in their motions. He also dangled an object from a thread and observed its motion as it swung to and fro. He rolled balls on the floor to see at which angle or on what kind of floor they rolled the furthest. On the basis of all these observations he concluded that Copernicus' heliocentric theory was correct. He published his findings in a book.

Figure 6.23: Galileo's portrait on the cover of one of his books. At the top are two figures, the one on the right looking through a telescope while the one on the left is using apparatus to measure

The church wasn't happy with the sun replacing the earth at the centre of the universe. It banned Galileo's book and put pressure on him to retract his observations. Galileo accepted the church's condition. But he continued to perform his experiments secretly, publishing his findings in countries where the Pope's writ did not extend.

Galileo's trial by the church became a symbol of the confrontation between religious dogma and science - with the upholders of religion suppressing scientific findings. It can also be seen as a confrontation between powerful vested interests on the one hand and the investigative spirit of the human beings on the other.

The English scientist Isaac Newton (born 1642, died 1727 CE) proposed his famous theory of gravity and the motion of astronomical bodies on the basis of the discoveries of Kepler and Galileo. Newton's work ushered in a new age where science was accorded the highest esteem.

We have seen how a new science was born in the Early-Modern era after being freed from the bonds of religious misconceptions. This new science was not based on old assumptions but upon experiment and mathematical proofs. The Age of the Renaissance began with the assumption that Aristotle's beliefs and methods were the proof of scientific knowledge. But with the onset of the scientific revolution, it was established that many of Aristotle's basic theories and methods of study and investigation were wrong.

The earth revolves around the sun – what are the problems a layman would face in believing this to be true?

6.3.1 Maritime Expeditions and Geographical Discoveries

At the time when Copernicus, Galileo and other scientists were conducting their investigations, sailors from Europe were crossing the oceans to find new sea lanes to new worlds. Before this, their voyages were mostly along the shallow coasts from port to port. They seldom ventured into the deep oceans. The main reason was that they did not have any maps of the seas which could help them find their way. There was only one way they could resolve the problem – observe the stars and planets and use their positions to find out where they were on the earth's surface and which path they should choose.

These mariners (sailors) set out for new lands in search of wealth. Many of them also wanted to spread Christianity in the new worlds they discovered to gain fame and salvation. The kings and emperors supported them in their search, especially because they wanted to find a new sea route to India and China. The mariners believed they could reach India if they sailed around the African continent. If the world was round and not flat as the scientists were then proposing, they could also sail across the western seas to reach the countries in the far-east. That meant they could cross the Atlantic Ocean to reach the eastern coast of China.

The Portuguese navigator (one who works out the route of ships on the sea) Vasco da Gama (born 1460, died 1524 CE) sailed around Africa to reach Calicut in the coastal Indian state of Kerala in 1498. Before him, the Italian navigator Christopher Columbus (born 1451, died 1506 CE) set out westwards to India, reaching a group of islands off the coast of a new continent. He thought he had reached India so he called the islands the 'Indies' and its inhabitants 'Indians'. Later, the Portuguese navigator Amerigo Vespucci (born 1454, died 1512) established this was a new continent by sailing around it. The continent was named America after him. A few years later, the Portuguese navigator Ferdinand Magellan (born 1480, died 1521 CE) decided to sail his ships around the entire world to prove the earth was round. He could not complete his circumnavigation, dying along the way, but his crew completed the journey.

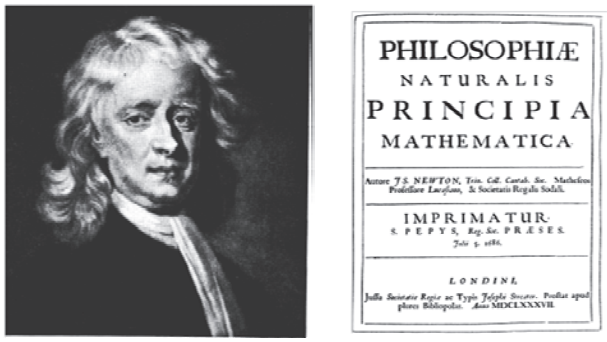

Figure 6.24: A 1741 portrait of Isaac Newton and (right) the cover of his monumental book 'Principia Mathematica' published in 1687

The mathematicians of Kerala (14th to 16th century)

The Indian state of Kerala had a robust mathematical tradition that flourished from the 14th to the 16th century. The major mathematicians of the time were Madhava, Neelakanta Somayaji, Parameshwara and Narayana Bhattathiri. They made several new discoveries in astronomy and mathematics. The most significant was the basic principles of calculus, which Isaac Newton later discovered in Europe and used to establish his theory of gravity. These mathematicians from Kerala were unable to develop their work and their discoveries where not propagated to the world.

EXERCISES

1. What were the factors that made the changes in the Middle Ages possible?
2. Why did kings establish autocratic monarchies? What role did the middle class play in their formation?
3. How did urbanisation and trade help in the development of the middle class?
4. In what way did trade and war contribute to the exchange of scientific knowledge between countries?
5. What were the unique features of ancient European classical literature? Why didn't the study of classics flourish during the Middle Ages?
6. What impact did the caste system have on the development of intellectual thought in India?
7. How did the scholars of Islamic countries contribute to the development of science in humanist Europe?
8. What were the similarities and differences between the lieterary studies of Humanists in Europe and the middle class in India?
9. How did Humanism challenge the authority of the church?
10. How were Renaissance paintings and sculpture different from the art of the Middle Ages?
11. What is the meaning of realism in art? Which Hindi films do you think are realistic? Explain with reasons.
12. Do you think Mughal paintings were realistic art? Give reasons.
13. What are the special features of Mughal paintings? Identify them in a painting before answering.
14. What is the contribution of Islamic architecture to Renaissance architecture?
15. What impression did the new Renaissance architecture seek to make on observers?
16. Scientific investigations were rarely undertaken in Europe and India during the Middle Ages. What could have been the reasons?
17. What was the contribution of China, India and the Arab countries to the development of science in Europe?
18. How was the importance of experiment, observation and calculation established in science? Can scientific knowledge progress without them?
19. Why did the European navigators set out on their maritime expeditions? Why was science important for their sea voyages?

PROJECT WORK

1. Make an album of Renaissance paintings. Write a couple of sentences on the artistic creations of each of the painters.
2. How were books made in India before the invention of the printing press? What were the problems in this process? Collect background information and some sample illustrations from these books.
3. Collect biographical information on the life and work of the famous humanists Erasmus and Machiavelli.
4. Find out details of the life of Galileo and his scientific investigations. Use this information to prepare an illustrated essay.

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